# Suzie DERMINON

Post-doctoral fellow · Ecology

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## Current position

**Post-doctoral position in Ecology and Crop production** — *Uppsala (Sweden)*

SLU - *Department of Ecology* — *January 2025 - January 2027 (planned)*

Referent researchers: Giulia VICO, Riccardo BOMMARCO
Title: Role of biodiversity and landscape for crop yield and its reliability under disturbance

## Professional Experience

**Research engineer** — *Gif-sur-Yvette (France)*

Université Paris Saclay - *Ecology, Systematic, Evolution Lab (ESE)* — *2020-2021 - nine months*

Project "Functionalities of terrestrial environments and effectiveness of management practices" (FONTE).
Bibliographical synthesis, interviews with municipal managers, and professional network animation.
Advisor: François CHIRON

**Ms2 internship** — *Paris (France)*

National Federation of Allotment and Community Gardens (FNJFC) — *2020 - six months*

Subject: Description of cultivated biodiversity and gardening practices in allotment and leisure gardens.
Advisors: Christine AUBRY, Badis MERAD

**Ms1 internship** — *Montpellier (France)*

Université de Montpellier - *MARine Biodiversity, Exploitation & Conservation lab (Marbec)* — *2019 - five months*

Subject: Modeling the carbon cycle of Pacific tuna populations and the impacts of human exploitation.
Advisor: David MOUILLOT

## Education

**PhD in Ecology** — *Gif-sur-Yvette (France)*

Université Paris Saclay - *Ecology, Systematic, Evolution Lab (ESE)* — *October 2021 - November 2024*

Advisors: François CHIRON, Audrey MURATET, Pierre-Alexis HERRAULT
Title: Influence of urbanisation on biodiversity: spatio-temporal dynamics of plants and birds in changing cities.
Defended: November 15th 2024
https://theses.hal.science/tel-04815124v1

**Master Degree in Engineering for Urban Green Spaces** — *Paris (France)*

Agro Paristech school — *2019 - 2020*

Technical, social and environmental approaches to the design and rehabilitation of sustainable urban spaces

**Master Degree in Engineering, department "Environmental science"** — *Palaiseau (France)*

Ecole polytechnique — *2016 - 2021*

Ecology, Fluid Mechanics & Climatology, Applied Mathematics, Computer Science & Modelling

## Skills

**Statistical Analysis:** Experienced in multiple linear regression, GLMs, GLMMs, GAMs, hierarchical partitioning, model ranking and selection, and model performance assessment

**Big Data Analysis:** Proficient with big datasets, especially from citizen science programs. Skilled in organizing, cleaning data, and addressing biases. Practiced in multidimensional analysis such as PCAs.

**Programming :** Expert in R, with experience in Python. `Tidyverse`, `dplyr` and `ggplot` enthusiast. Adept at learning new languages for specific tasks, including JavaScript for the *Google Earth Engine* API.

**Spatial Data Handling & Analysis:** Skilled in QGis and in spatial analysis with R (`SF` and `Raster` packages). Comfortable with vector and raster data, experienced in computing vegetation indices like NDVI.

**Spatio-Temporal Approach of Ecology:** Strong background in temporal ecological studies. Experienced in describing landscape through temporal trajectories, and in handling multicollinearity in data.

## Publications

### Published

**Derminon, S.**, Chiron, F., Muratet, A., Herrault P.A. 2024. Textural indices of aerial images to assess the effects of urban landscapes history on plant and bird communities. Ecological Indicators, 62, pp.112002. DOI: 10.1016/j.ecolind.2024.112002

Mouillot, D., **Derminon, S.**, Mariani, G., Senina, I., Fromentin, J.M., Lehodey., P., Trousselier, M. 2023. Industrial fisheries have reversed the carbon sequestration by tuna carcasses into emissions. Global Change Biology, 29, 5062–5074 DOI: 10.1111/gcb.16823

### In Prep

**Derminon, S.**, Muratet, A., Herrault, P.A., Chiron, F. "Delayed ecological responses to urbanization history in bird and plant communities". *Under reviwe*. Preprint: DOI: 10.22541/au.175187890.01780136/v1

***Delsanti, V.***, Philippe, M., **Derminon, S.** "Urban tolerance of common French bird species: how distance to the closest building shapes bird communities". *In prep*.

## Presentations

### Talks

Winter 2025. *Time in urban ecology: Effect of urban trajectories on bird and plant population trends* . Swedish Oikos Meeting, Stockholm, Sweden.

Summer 2024. *Time travel in landscape ecology: Archive aerial images and texture-based indices reveal past urban landscapes and their effects on plants and birds.*. Contributed session, European Congress of Conservation Biology, Bologna, Italy.

Winter 2022. *The texture of aerial archives to understand the effects of history on plant and bird communities in urban environments.*. Geography seminar, Image City Environment Laboratory (LIVE), Strasbourg, France.

### Posters

Summer 2022. *Back to the future: are historical aerial photos the Delorean of ecology?*. Demain : Nature et ville ?, INSA Centre-Val-de-Loire, Blois, France.

Autumn 2024. *Time in urban ecology: Effect of urban trajectories on bird and plant population trends* . SFE2, Lyon, France.

## Teaching & Scientific mediation Experience

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| Spring 2024 | **MS1 "Earth Observation and Geomatics"**, Seminar supervisor - 12h<br>Image, City, Environment Lab (LIVE) - *Université de Strasbourg* |
| 2021 - 2023 | **Weekly scientific mediation workshops with schoolchildren (7 to 15 years-old)** , Science facilitator - 64 days<br>Science Initiation and Awareness House (MISS) - *Université Paris-Saclay* |

## Mentoring

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| 2023 | **Vincent Delsanti**, Ms2 student<br>Ecole Centrale-Supélec - *Six months* |

## Commitment & Professional Development

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| 2025-today | **Vice-president**, SLU Post-doc Association (SLUPA) |
| 2023-2024 | **Non-permanent representative**, Ecology, Systematic, Evolution lab (ESE) |
| Spring 2024 | **Organizing member**, Workshop: "Young researchers in urban ecology" |
| 2021-2024 | **PhD. students representative**, Graduate school- *Plant Science: from gene to ecosystem* |
| 2021-2024 | **Commitment to making science accessible to the general public**, Various institutions |